

LUCIDMSK CME PROGRAM

MODULE 07 — POST-TEST

Ankle & Foot MRI: Ligaments, Tendons & Fracture Classification

Ankle · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)[™]

Read each question carefully and select the single best answer. A score of $\geq$ **70%** (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 Sanders Type III calcaneal fracture on CT has:

- A Non-displaced fragments
- B Two-part posterior facet; one fracture line
- C Three-part posterior facet; two fracture lines
- D Comminuted; >3 fragments

2 ATFL Grade III tear on MRI:

- A Edema; fibers intact
- B Partial fiber discontinuity; attenuated
- C Complete discontinuity; wavy/absent; fluid gap
- D Chronic thickening only

3 Most common location for peroneus brevis longitudinal split tear:

- A Cuboid notch
- B Proximal to the ankle
- C Retrofibular groove
- D Adjacent to peroneal tubercle

4 Achilles tendon AP diameter 10 mm on sagittal MRI with intratendinous T2 signal. Interpretation:

- A Normal — up to 12 mm is OK
- B Fusiform tendinopathy (>6 mm threshold); mucinoid degeneration
- C Partial tear requiring surgery
- D Complete rupture

5 Turf toe Grade III on MRI:

- A Plantar plate sprain; fibers intact
- B Partial plantar plate tear; sprung MTP
- C Complete plantar plate disruption $\pm$ sesamoid fracture $\pm$ MTP instability
- D Normal MRI; clinical diagnosis

6 Eichenholtz Stage I Charcot arthropathy on radiograph shows:

- A Normal X-ray; clinical swelling only
- B Fragmentation — cortical disruption; subluxation; debris
- C Coalescence — decreased debris; bone repair
- D Reconstruction — remodeling; stable

7 Spring ligament (superomedial calcaneonavicular) tear. Most clinically relevant associated finding:

- A Peroneus brevis split
- B Posterior tibialis tendon (PTT) dysfunction
- C ATFL tear
- D Sinus tarsi syndrome

8 Weber B ankle fracture (Danis-Weber). Location:

- A Below the syndesmosis
- B AT the level of the syndesmosis
- C Above the syndesmosis
- D Isolated medial malleolus

9 Sinus tarsi syndrome on MRI:

- A Fluid in sinus tarsi only

- | | |
|----------|---|
| B | Loss of normal fat signal; T1 low/T2 high signal; loss of cervical and interosseous ligaments |
| C | Peroneal tendon tear |
| D | Calcaneal stress fracture |

10 Freiberg infraction Smillie Stage III:

- | | |
|----------|--|
| A | Normal X-ray; MRI subchondral edema |
| B | Widening/flattening beginning; no fragmentation |
| C | Central depression; loose body formation |
| D | Complete disintegration; secondary MTP arthrosis |

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	C	C	C	B	C	B	B	B	B	C

Q1 Sanders Type III calcaneal fracture on CT has: **Answer: C**

- A Non-displaced fragments
- B Two-part posterior facet; one fracture line

✓ **Three-part posterior facet; two fracture lines**

- D Comminuted; >3 fragments

EXPLANATION Sanders III = three-part posterior facet (two vertical fracture lines through the posterior facet). Type II = two-part (one line). Type IV = comminuted (>3 fragments). Type I = non-displaced regardless of lines.

Q2 ATFL Grade III tear on MRI: **Answer: C**

- A Edema; fibers intact
- B Partial fiber discontinuity; attenuated

✓ **Complete discontinuity; wavy/absent; fluid gap**

- D Chronic thickening only

EXPLANATION Grade III = complete tear: fibers absent/discontinued with fluid gap. Grade II = partial discontinuity. Grade I = edema with intact fibers. Always assess CFL and PTFL concurrently.

Q3 Most common location for peroneus brevis longitudinal split tear: **Answer: C**

- A Cuboid notch
- B Proximal to the ankle

✓ **Retrofibular groove**

- D Adjacent to peroneal tubercle

EXPLANATION Peroneus brevis split tears occur most commonly at the retrofibular groove, where it is compressed against the fibula during inversion. The "C-shape" or crescent appearance on axial MRI is characteristic.

Q4 Achilles tendon AP diameter 10 mm on sagittal MRI with intratendinous T2 signal. Interpretation: **Answer: B**

- A Normal — up to 12 mm is OK

✓ **Fusiform tendinopathy (>6 mm threshold); mucinoid degeneration**

- C Partial tear requiring surgery
- D Complete rupture

EXPLANATION Achilles AP diameter >6 mm = abnormal tendinopathy. Intratendinous T2 signal = mucinoid degeneration. Partial tear = focal high T2 with fiber discontinuity. Complete rupture = full-thickness gap with retraction.

Q5 Turf toe Grade III on MRI:

Answer: C

- A Plantar plate sprain; fibers intact
- B Partial plantar plate tear; sprung MTP

✓ **Complete plantar plate disruption ± sesamoid fracture ± MTP instability**

- D Normal MRI; clinical diagnosis

EXPLANATION Turf toe Grade III = complete plantar plate disruption ± sesamoid fracture ± 1st MTP instability/dislocation. Grade I = sprain. Grade II = partial tear. Grade III may require surgical assessment.

Q6 Eichenholtz Stage I Charcot arthropathy on radiograph shows:

Answer: B

- A Normal X-ray; clinical swelling only
- ✓ **Fragmentation — cortical disruption; subluxation; debris**
- C Coalescence — decreased debris; bone repair
- D Reconstruction — remodeling; stable

EXPLANATION Stage I (Fragmentation) = active breakdown: cortical disruption, fracture, joint subluxation, and periarticular bone debris. Stage 0 (Sanders) = prodromal; normal X-ray. Stage II = coalescence (repair). Stage III = reconstruction.

Q7 Spring ligament (superomedial calcaneonavicular) tear. Most clinically relevant associated finding:

Answer: B

- A Peroneus brevis split
- ✓ **Posterior tibialis tendon (PTT) dysfunction**
- C ATFL tear
- D Sinus tarsi syndrome

EXPLANATION Spring ligament tears are strongly associated with PTT dysfunction and adult acquired flatfoot deformity. The spring ligament is the "sling" supporting the talar head — its failure causes medial arch collapse.

Q8 Weber B ankle fracture (Danis-Weber). Location:

Answer: B

- A Below the syndesmosis
- ✓ **AT the level of the syndesmosis**
- C Above the syndesmosis
- D Isolated medial malleolus

EXPLANATION Weber B = fibula fracture AT the level of the syndesmosis. Weber A = below (stable). Weber C = above (Maisonneuve; syndesmosis disruption; unstable). Stability depends on B subtype and medial column integrity.

Q9**Sinus tarsi syndrome on MRI:****Answer: B**

A Fluid in sinus tarsi only

✓ **Loss of normal fat signal; T1 low/T2 high signal; loss of cervical and interosseous ligaments**

C Peroneal tendon tear

D Calcaneal stress fracture

EXPLANATION Sinus tarsi syndrome = loss of normal fat (T1 low, replaced by fibrosis or ganglion), loss of the cervical and interosseous talocalcaneal ligaments. Clinically: lateral hindfoot pain after inversion injury.

Q10**Freiberg infraction Smillie Stage III:****Answer: C**

A Normal X-ray; MRI subchondral edema

B Widening/flattening beginning; no fragmentation

✓ **Central depression; loose body formation**

D Complete disintegration; secondary MTP arthrosis

EXPLANATION Smillie Stage III = central depression/collapse of the metatarsal head with loose body formation. Stage I = subchondral changes only. Stage II = flattening without fragmentation. Stage IV–V = progressive disintegration and arthrosis.